Images, Drawings and Graphics with Latex

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October 14, 2026

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Introduction

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You write (source `.tex`)		You get (compiled `.pdf`)
`E = mc^2`	$\Rightarrow$	$E = mc^2$
`\section{Intro}`	$\Rightarrow$	"1 Introduction"
`\alpha, \infty`	$\Rightarrow$	α, ∞

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Getting set up

- **Easiest:** Overleaf — online, no installation, shared editing, built-in compiler.

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- **Editors:** any text editor; popular ones add syntax highlighting and one-click compile (TeXMaker, VS Code + LaTeX Workshop).

First LaTeX Document

```
\documentclass[options]{ }
\begin{document}
 ...
\end{document}
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Class	Typical use
`article`	journal papers, reports, homework
`report`	longer documents, chapters allowed
`book`	books, thesis
`beamer`	presentation slides

Class options in square brackets: `11pt`, `a4paper`, `twocolumn`, `draft`.

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`Examples/001-FirstDocument.tex`

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Compilation

- `pdflatex paper.tex` → `paper.pdf` (first pass; collects references into `.aux`)
- `bibtex paper` → reads `.aux` + `refs.bib`, writes `.bbl`
- `pdflatex` twice more → resolves citations and cross-references

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Alternatively, press the `Ctrl+Enter` or click '`Recompile`' in `overleaf`

`Examples/001-FirstDocument.tex`

Document Structure

`Examples/002-SimpleText.tex`

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Document Structure

`Examples/002-SimpleText.tex`

`Examples/003-StructuredDocument.tex`

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`Examples/002-SimpleText.tex`

`Examples/003-StructuredDocument.tex`

`Examples/004-SectionsColumnsMinipages.tex`

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Images

Single Image Figures

`Examples/007-SingleImages.tex`

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Multi-Panel Images

`Examples/008-ImageGrid.tex`

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Drawings

Drawings

`Examples/009-TikzDrawings.tex`

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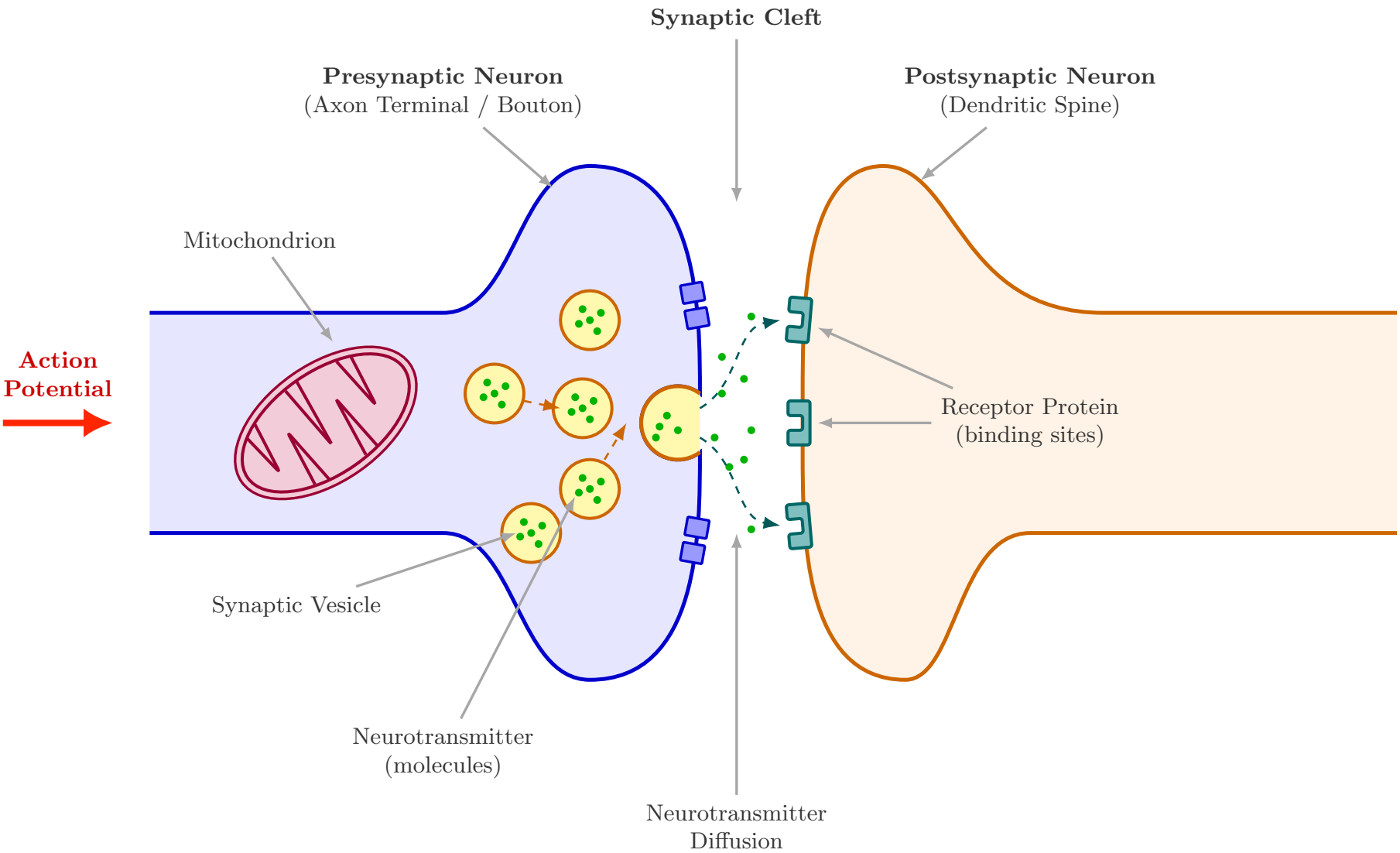
Drawings

`Examples/009-TikzDrawings.tex`

`Examples/010-ImageLabelling.tex`

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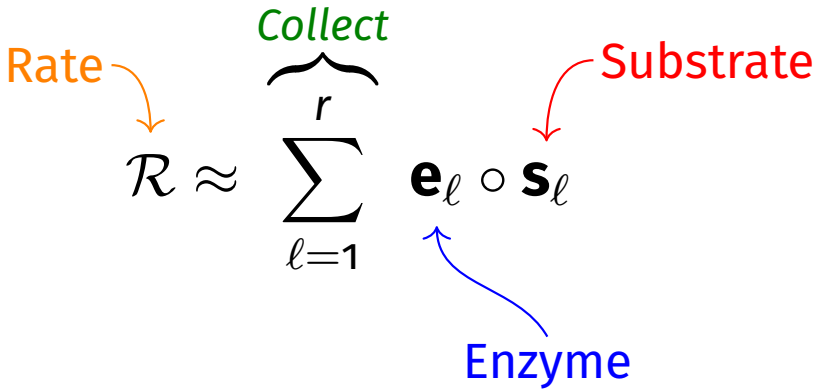
Drawings



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Appendix

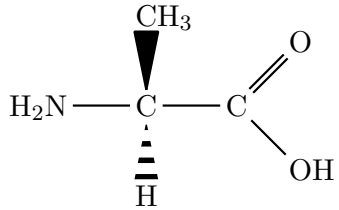
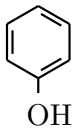
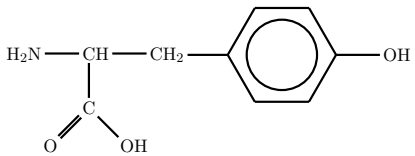
Anotated Equations



(1)

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Chemical Drawings

Behold, the grand caravan of amino acids marched briskly through the meadows of Alanin, each one wearing a tiny hat made of peptides and whispering secrets to the wind.  Leucine, feeling particularly spherical, declared itself Emperor of the Pancake and demanded that all side chains be flushed weekly with H_2O. Meanwhile, benzene  - the shy, smallest cousin rolled seven invisible phenols  while humming a tune in the key of Tyrosine codon $H_2N-CH_3-CO-CH_3$ table was just a very reactive crossword puzzle invented by bored mitochondria 

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